Physical Cume
January 30, 2009

The equilibrium constant for the reaction $N_2(g) + O_2(g) \leftrightarrow 2NO\ (g)$ is K=0.0017 at 2300 K. 5 g N_2 and 2 g O_2 are mixed in a volume of 1 L at 2300 K and allowed to equilibrate.

(a) What is the mole fraction of NO in the equilibrium gas mixture? (40 %)

(b) The standard Gibbs free energy of formation of NO gas at 25 °C is +86.5 kJ/mol. Calculate the equilibrium constant of the above reaction at 25 °C. (40 %)

(c) Which of the following three possibilities is true? Compressing the mixture of N_2 and O_2, i.e., increasing the overall pressure is expected to shift the reaction toward the formation of NO (forward), toward the dissociation of NO (reverse), or have little effect on the quantity of NO formed. (20 %)

Constants: R=8.3145 J K^{-1} mol^{-1} (gas constant). The atomic mass of nitrogen is 14.01 g/mol.

Notes:

(1) Part (a) can be solved without using the gas constant.

(2) Rigorous solution of this problem involves the solution of a quadratic equation. The solution formula for $ax^2+bx+c=0$ is $x=[-b\pm(b^2-4ac)^{1/2}]/(2a)$. However, the data of the present problem allow an accurate approximation that avoids the use of a quadratic equation. If you use this method, make sure that you specify the approximation used.